



thuRsdays

LECTURE 3- WHAT IS THE
STORY OF YOUR DATA

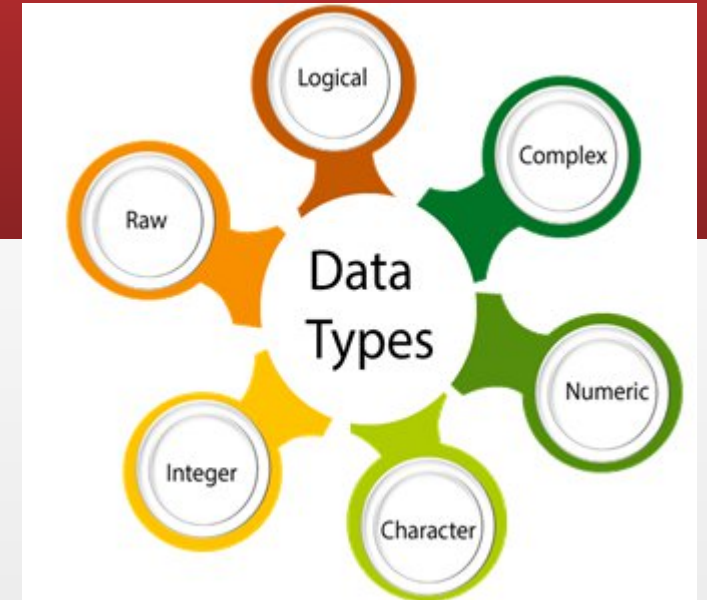


Recap

- What is R
 - Statistical and mathematical programming environment
 - Free and Open Source Software
- You can access R in many ways
 - Command line
 - Most challenging
 - Graphical User Interface (GUI)
 - RGUI (Windows)
 - Interactive Development Environment (IDE)
 - RStudio, Jupyter Lab
 - GUI based R Analysis tools
 - No Code Exposed - Jamovi, RKWard
 - Code Exposed - RCommander
 - Limited to Graphical Analysis

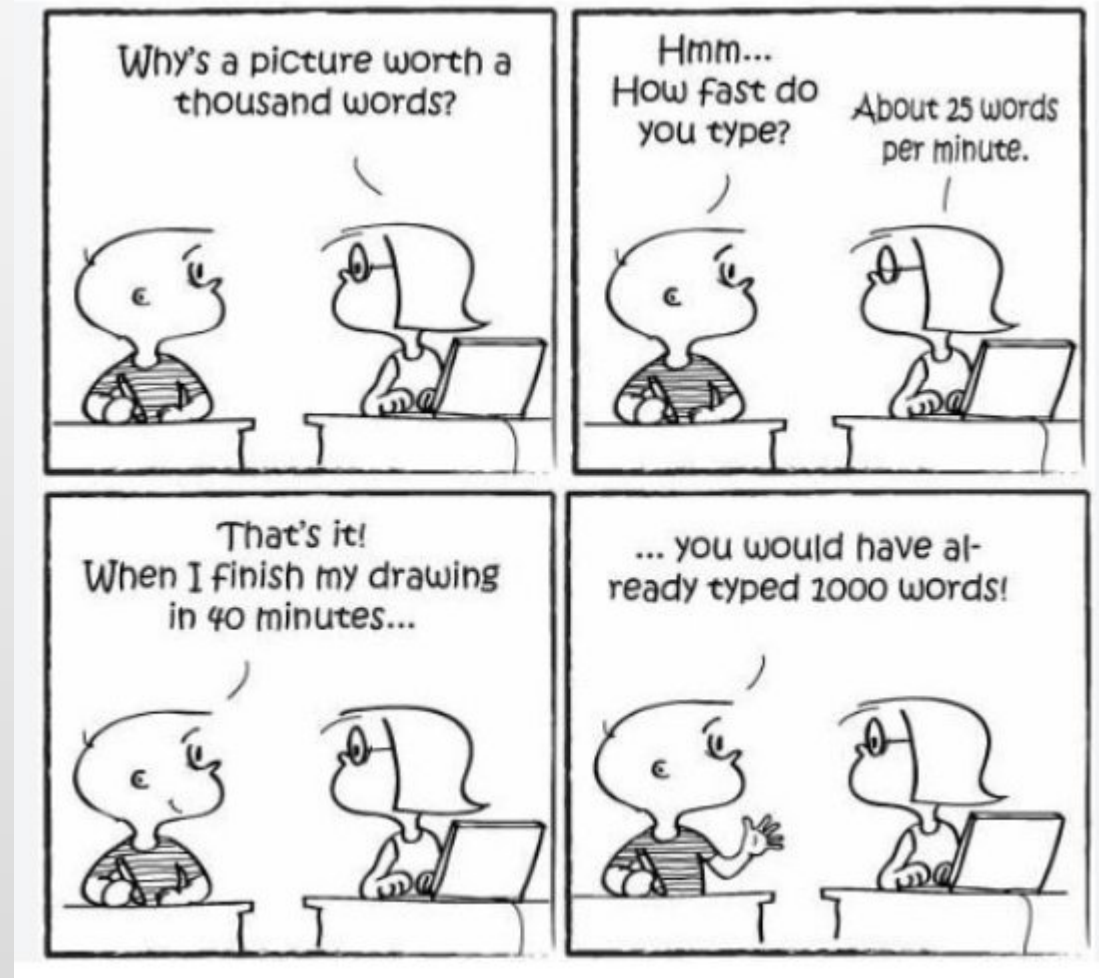
You can use Posit Cloud to run RStudio and R

<https://posit.cloud/>



Visualization using R

- **Storytelling with Data**
 - Data Analysis + Visualizations + Structured Narrative
- Present complex information to drive action
- R can help you with
 - Data Analysis
 - Visualization



Visualization Approaches in R

- R offers three approaches to visualization
 - Base R Graphics
 - Primarily procedural
 - Uses functions like plot, boxplot, barplot,...
 - We create a plot and add elements
 - Labels, axes, title, range (limits)
 - Lightweight, flexible and best approach for simple layouts
 - Lattice Graphics
 - Useful for multivariate conditional plot
 - How does the relationship vary across a set of conditions
 - Best when you have a handful of factors to condition upon
 - Still procedural
 - xyplot(), bwplot(), histogram(), densityplot(), levelplot(), wireframe()
 - Grammar of Graphics (ggplot)
 - Graph = Data + Aesthetics + Geometry + Statistics + Scales + Facets
 - High Publication quality graphs
 - Steep learning curve of the grammar
 - May require other packages

Spatial Visualization

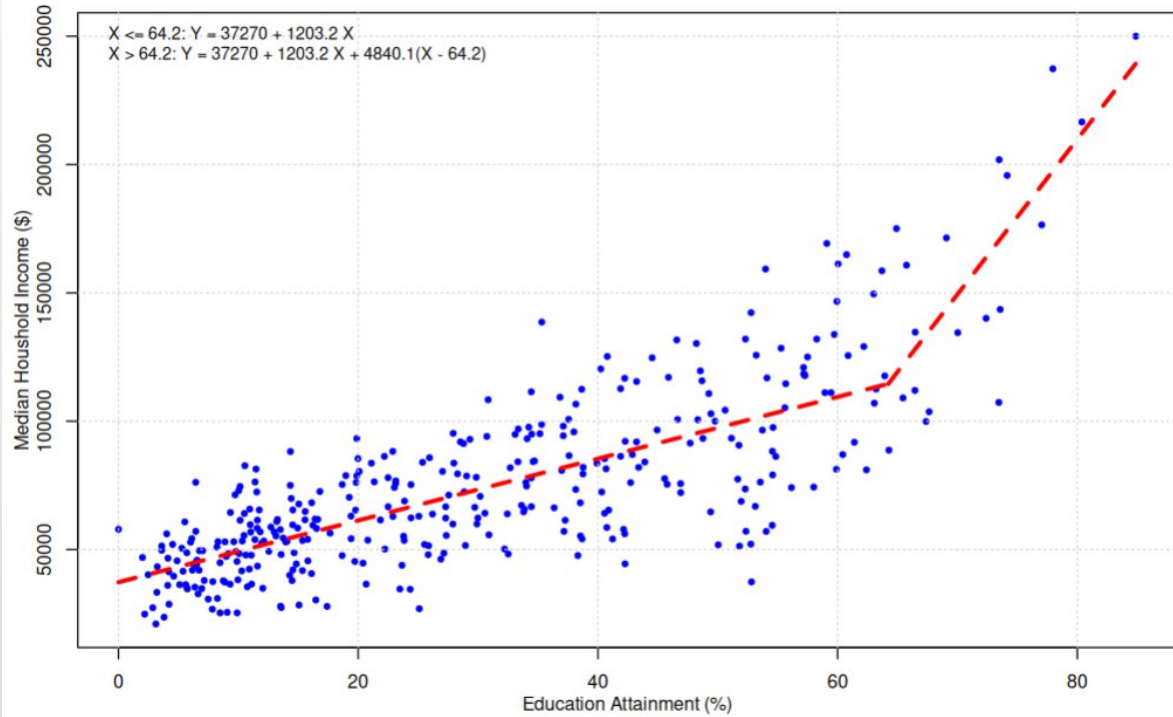
Spatio-temporal Visualization

Interactivity

Scientific Visualization in higher dimensions

Hands-On Examples

Median Household Income vs Educational Attainment (%)



San Antonio Airport

Daily Total Precipitation

